

The Euclidean space $\mathbb{R}^m$ (vector space)

24.11.2025
Lecture 8
(not mid-term)

• $x = (x_1, \dots, x_m) \in \mathbb{R}^m$

Matrix notation: $x = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}_{m \times 1}$

• $x+y = (x_1+y_1, \dots, x_m+y_m)$

• $\alpha x = (\alpha x_1, \dots, \alpha x_m), \alpha \in \mathbb{R}$

Amap $\langle \cdot, \cdot \rangle : \mathbb{R}^m \times \mathbb{R}^m \rightarrow \mathbb{R} =$ a scalar product (inner product)

- Def.
1. $\langle x, y \rangle = \langle y, x \rangle, (\forall) x, y \in \mathbb{R}^m$
 2. $\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle, \dots$
 3. $\langle x, x \rangle \geq 0, (\forall) x \in \mathbb{R}^m \setminus \{0\}$

Def: The dot product $\rightarrow$ the most important scalar product.

$$x \cdot y := x_1 y_1 + \dots + x_m y_m, (\forall) x, y \in \mathbb{R}^m$$

In matrix notation:

$$x \cdot y = x^T y = [x_1 \dots x_m]_{1 \times m} \cdot \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}_{m \times 1} = x_1 y_1 + \dots + x_m y_m$$

Def: Two vectors $x, y \in \mathbb{R}^m$ are PERPENDICULAR (orthogonal)

$$x \cdot y = 0$$

Ex: $M \in \mathbb{R}^{m \times m}$
 $M^T = M$ (symmetric)

$x^T M x > 0$ (positive definite matrix), $(\forall) x \in \mathbb{R}^m$

$$\text{Let } \langle x, y \rangle := x^T M y = [x_1 \dots x_m]_{1 \times m} \begin{bmatrix} m_{11} & m_{1m} \\ \vdots & \vdots \\ m_{m1} & m_{mm} \end{bmatrix}_{m \times m} \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}_{m \times 1}$$

$$= [\quad]_{1 \times 1}$$

check a) $\langle x, y \rangle \stackrel{?}{=} \langle y, x \rangle$

$$(AB)^T = B^T \cdot A^T$$

$$x^T M y = (x^T M y)^T = y^T \cdot (x^T M)^T = y^T \cdot M^T \cdot x = y^T M x = \langle y, z \rangle.$$

b) $\langle \alpha x + \beta y, z \rangle \stackrel{?}{=} \alpha \langle x, z \rangle + \beta \langle y, z \rangle, \quad (\forall) \alpha, \beta \in \mathbb{R}$
 $(\forall) x, y, z \in \mathbb{R}^m$

$$\begin{aligned} (\alpha x + \beta y)^T M z &= \\ &= \alpha x^T M z + \beta y^T M z \\ &= \alpha \langle x, z \rangle + \beta \langle y, z \rangle \end{aligned}$$

c) $\langle x, x \rangle = x^T M x > 0, \quad (\forall) x \in \mathbb{R}^m \setminus \{0_m\}$

Def: NORM $\approx$ length of a vector.

A function $\|\cdot\|: \mathbb{R}^m \rightarrow [0, \infty)$ is called a norm if:

1) $\|x\| = 0 \Leftrightarrow x = 0$.

2) $\|\alpha x\| = |\alpha| \cdot \|x\|, \quad (\forall) x \in \mathbb{R}^m, \alpha \in \mathbb{R}.$

3) $\|x+y\| \leq \|x\| + \|y\|, \quad (\forall) x, y \in \mathbb{R}^m$ (Δ ineq.)

! Any scalar product generates a norm on $\mathbb{R}^m$:

$$\|x\| = \sqrt{\langle x, x \rangle}$$

↳ Proof: (that $\|x\| = \sqrt{\langle x, x \rangle}$ is a norm)

$$\sqrt{\langle x, x \rangle} = \|x\|$$

• $\|x\| = 0 \Leftrightarrow \langle x, x \rangle = 0 \Leftrightarrow x = 0_m$. (the vector with no comp)

(Recall that $\langle x, x \rangle = 0, (\forall) x \in \mathbb{R}^m \setminus \{0_m\}$) ?

• $\|\alpha x\| = \sqrt{\langle \alpha x, \alpha x \rangle} = \sqrt{\alpha^2 \langle x, x \rangle} = \sqrt{\alpha^2} \cdot \sqrt{\langle x, x \rangle} = |\alpha| \cdot \|x\|,$
 $(\forall) \alpha \in \mathbb{R}, (\forall) x \in \mathbb{R}^m.$

• $\|x+y\| \stackrel{?}{\leq} \|x\| + \|y\|$

$$\sqrt{\langle x+y, x+y \rangle} \stackrel{?}{\leq} \sqrt{\langle x, x \rangle} + \sqrt{\langle y, y \rangle}$$

$$\begin{aligned}
 \sqrt{\langle x+y, x+y \rangle} &= \sqrt{\langle x, x \rangle + \underbrace{\langle x, y \rangle}_{= \langle y, x \rangle} + \langle y, x \rangle + \langle y, y \rangle} \\
 &= \sqrt{\langle x, x \rangle + \langle y, y \rangle + 2 \langle x, y \rangle} \\
 &= \sqrt{\|x\|^2 + 2 \langle x, y \rangle + \|y\|^2}
 \end{aligned}$$

? If $\langle x, y \rangle \leq \|x\| \cdot \|y\|$, then

$$\leq 2\|x\| \cdot \|y\|$$

$$\leq \sqrt{(\|x\| + \|y\|)^2} = \|x\| + \|y\|$$

Theorem (Cauchy Schwarz imag.)

$$\text{Let } x, y \in \mathbb{R}^n \Rightarrow |\langle x, y \rangle| \leq \|x\| \cdot \|y\|.$$

↳ Proof:

$$\text{Let } f: \mathbb{R} \rightarrow \mathbb{R}, f(t) = \langle x+ty, x+ty \rangle.$$

Notice that $f(t) \geq 0$ (from the def of the scalar prod.)

$$\text{But } f(t) = \langle x, x+ty \rangle + \langle ty, x+ty \rangle$$

$$= \langle x, x \rangle + t \langle x, y \rangle + t \langle y, x \rangle + t^2 \langle y, y \rangle$$

$$= \underbrace{\|x\|^2}_{\rightarrow c} + 2t \langle x, y \rangle + t^2 \underbrace{\|y\|^2}_{\rightarrow a}$$

Quadratic function in t .

$$\text{Since } f(t) \geq 0 \Rightarrow \Delta \leq 0.$$

$$\Delta = b^2 - 4ac = 4 \langle x, y \rangle^2 - 4 \|x\|^2 \|y\|^2 \leq 0$$

$$\Rightarrow 4 \langle x, y \rangle^2 \leq 4 \|x\|^2 \|y\|^2$$

$$\Rightarrow \langle x, y \rangle^2 \leq \|x\|^2 \|y\|^2$$

$$\Rightarrow |\langle x, y \rangle| \leq \|x\| \cdot \|y\|.$$

$$\Rightarrow -\|x\| \cdot \|y\| \leq \langle x, y \rangle \leq \|x\| \cdot \|y\|$$

$$|x_1 y_1 + \dots + x_n y_n| \leq \sqrt{x_1^2 + \dots + x_n^2} \cdot \sqrt{y_1^2 + \dots + y_n^2}$$

$$\Rightarrow (x_1 y_1 + \dots + x_n y_n)^2 \leq (x_1^2 + \dots + x_n^2) \cdot (y_1^2 + \dots + y_n^2)$$

Def: The euclidean norm: (generated by the dot prod.)

$$\|x\| = \sqrt{x \cdot x} = \sqrt{x_1^2 + \dots + x_n^2}$$

Theorem

$$n \in \{2, 3\} \quad (2D/3D)$$

$$x \cdot y = \|x\| \cdot \|y\| \cdot \cos \angle(x, y)$$

(the dot product of x, y has a geometrical meaning)

Other examples

1. $\|x\|_1 = |x_1| + \dots + |x_n|$ is a norm

↳ "1 norm"

2. $\|x\|_p = (|x_1|^p + \dots + |x_n|^p)^{1/p}$, $p \geq 1$

↳ "p norm"

3. $\|x\|_\infty = \max\{|x_1|, \dots, |x_n|\}$ norm

↳ "infinity norm"

Def: DISTANCE (METRIC)

A function $d: \mathbb{R}^n \times \mathbb{R}^n \rightarrow [0, \infty)$ is a distance if:

1) $d(x, y) = 0 \Leftrightarrow x = y$

2) $d(x, y) = d(y, x)$, $(\forall) x, y \in \mathbb{R}^n$

3) $d(x, z) \leq d(x, y) + d(y, z)$, $(\forall) x, y, z \in \mathbb{R}^n$ (triangle inequality)

! Any norm generates a distance on $\mathbb{R}^n$ given by

$$d(x, y) = \|x - y\|$$

Def: The Euclidean distance:

$$d(x, y) = \|x - y\| = \sqrt{(x_1 - y_1)^2 + \dots + (x_n - y_n)^2}$$

$$\left(\begin{array}{l} d(A, B) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2} \\ A = (x_1, y_1) \\ B = (x_2, y_2) \end{array} \right)$$

In case of a collection of vectors:

Def: **BOUNDED SET**

A set $A \subseteq \mathbb{R}^m$ bounded if: $\exists \kappa > 0$ s.t. $\|x\| \leq \kappa, (\forall) x \in A$.

Def: **OPEN BALL**. (open disc)

Let $x_0 \in \mathbb{R}^m, \kappa > 0$.

$$B(x_0, \kappa) := \{x \in \mathbb{R}^m \mid \|x - x_0\| < \kappa\}$$

measured with euclidean norm.

Def: **CLOSED BALL**: (closed disc $\rightarrow$ contains the circle)

$$\bar{B}(x_0, \kappa) := \{x \in \mathbb{R}^m \mid \|x - x_0\| \leq \kappa\}$$

Def: **NEIGHBOURHOOD**.

A set $V \subseteq \mathbb{R}^m$ is a n. of $x \in \mathbb{R}^m$ if:

$$\exists \kappa > 0 \text{ s.t. } B(x, \kappa) \subseteq V.$$

The neighbourhoods of x : $\mathcal{V}(x) := \{V \subseteq \mathbb{R}^m \mid V \text{ n. of } x\}$.

Interior ... $V \subseteq A$

Closure ... $V \cap A \neq \emptyset$

Boundary ... $V \cap A \neq \emptyset$

$$V \cap A^c \neq \emptyset$$

of a set A

the complement of a SET.

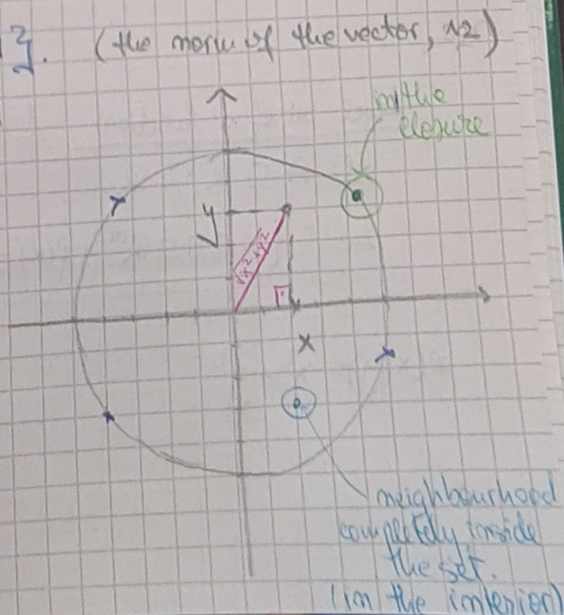
Ex: $A = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 < 1\}$. (the norm of the vector, $\sqrt{x^2 + y^2}$)

$$\text{int } A = A.$$

$$\text{cl } A = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$$

$$\text{bd } A = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$$

$\hookrightarrow$ toute pt. de pe cerc (both in A , and in A^c)



Def:

$A = \text{int } A \Rightarrow A$ is OPEN.

$A = \text{cl } A \Rightarrow A$ is CLOSED.

! $(\forall) A \subseteq \mathbb{R}^m, \text{int}(A) \subseteq A \subseteq \text{cl}(A).$

The following statements:

✓ the complement of an open set $\rightarrow$ CLOSED.

✓ the complement of a closed set $\rightarrow$ OPEN.

Sequences in $\mathbb{R}^m$.

A seq. (x^k) in $\mathbb{R}^m$ has vector elements $x^1, x^2, \dots, x^k$.

Def: (x^k) converges to $x \in \mathbb{R}^m$ if $\lim_{k \rightarrow \infty} \|x^k - x\| = 0$
 $\downarrow$
(the k -th vector)

$$\lim_{k \rightarrow \infty} x^k = x.$$

$$\text{EX: } x^k = \left(\frac{1}{k}, \frac{k}{k+1} \right)$$
$$\lim_{k \rightarrow \infty} x^k = (0, 1)$$

Theorem (each component converges to each comp.)

? A seq. x^k converges $\dots x_i^k \dots$

Level set of a function

$$f: A \subseteq \mathbb{R}^m \rightarrow \mathbb{R}.$$

$L_c := \{x \in A \subseteq \mathbb{R}^m \mid f(x) = c\}$ height is constant

"curba de nivel"

$$f(x, y) = x^2 + y^2$$

↓
slice

↓
1D
↓
2D